```
import pandas as pd

# Load the dataset
df_insurance = pd.read_csv('/content/insurance.csv')

print('*** Dataset Shape ***')
print(df_insurance.shape)
print('\n')

print('*** Column Names ***')
print(df_insurance.columns.tolist())
print('\n')

print('*** Data Types ***')
print(df_insurance.info())
print('\n')

print('*** First 5 Rows ***')
display(df_insurance.head())
```

```
*** Dataset Shape ***
(1338, 7)


*** Column Names ***
['age', 'sex', 'bmi', 'children', 'smoker', 'region', 'charges']


*** Data Types ***
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1338 entries, 0 to 1337
Data columns (total 7 columns):
 #   Column    Non-Null Count  Dtype
---  ------    --------------  -----
 0   age       1338 non-null   int64
 1   sex       1338 non-null   object
 2   bmi       1338 non-null   float64
 3   children  1338 non-null   int64
 4   smoker    1338 non-null   object
 5   region    1338 non-null   object
 6   charges   1338 non-null   float64
dtypes: float64(2), int64(2), object(3)
memory usage: 73.3+ KB
None


*** First 5 Rows ***
```

	age	sex	bmi	children	smoker	region	charges
0	19	female	27.900	0	yes	southwest	16884.92400
1	18	male	33.770	1	no	southeast	1725.55230
2	28	male	33.000	3	no	southeast	4449.46200
3	33	male	22.705	0	no	northwest	21984.47061
4	32	male	28.880	0	no	northwest	3866.85520

```
# Check for missing values
print('*** Missing Values ***')
print(df_insurance.isnull().sum())
print('\n')
```

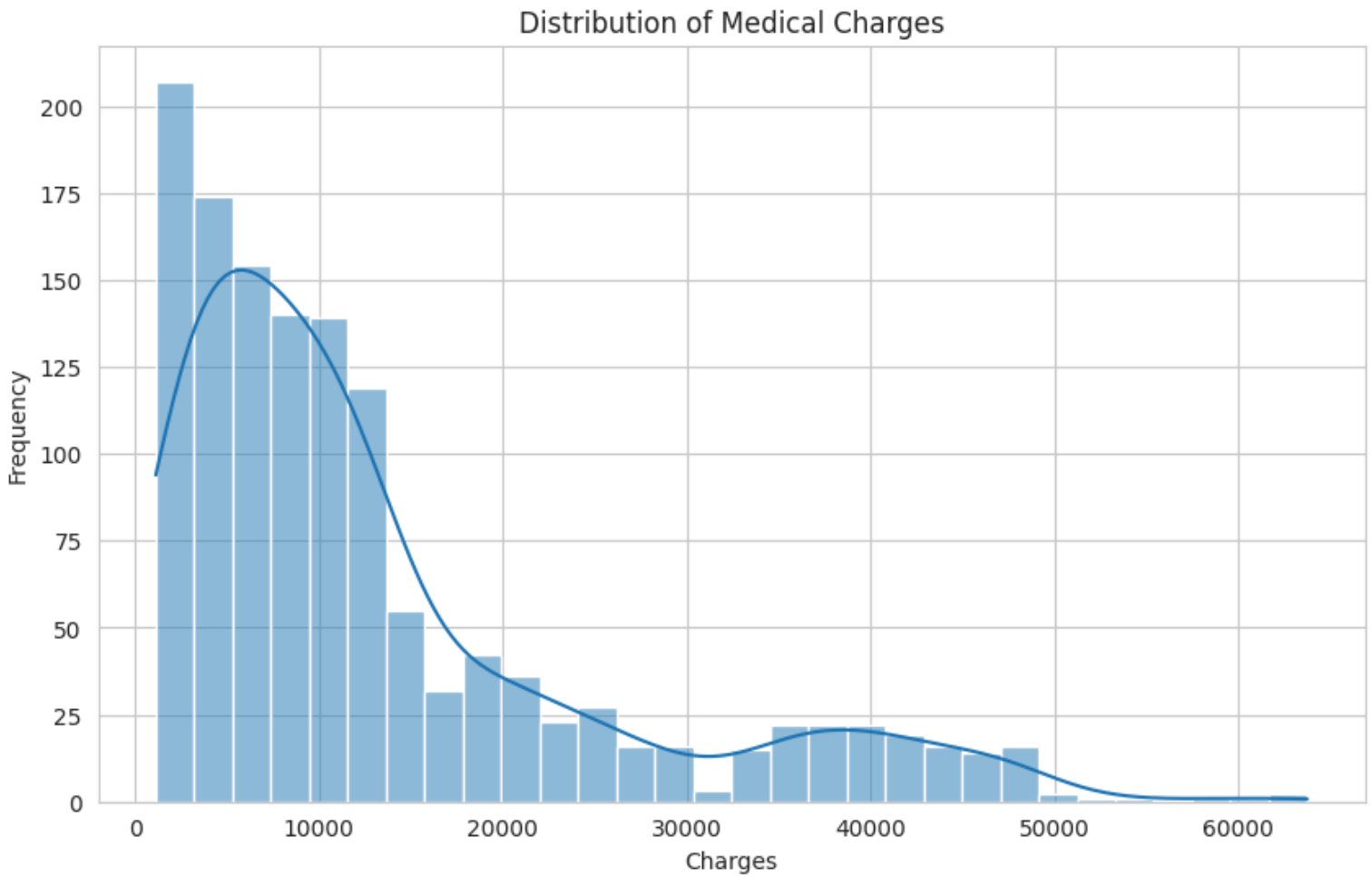
```
# Check for duplicate rows
print('*** Duplicate Rows ***')
print(df_insurance.duplicated().sum())
```

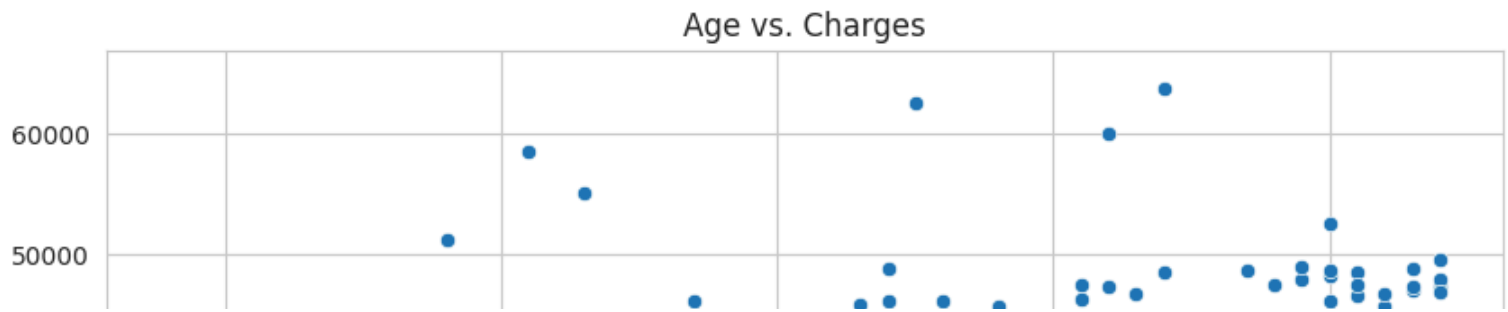
```
*** Missing Values ***
age         0
sex         0
bmi         0
children    0
smoker      0
region      0
charges     0
dtype: int64


*** Duplicate Rows ***
1
```

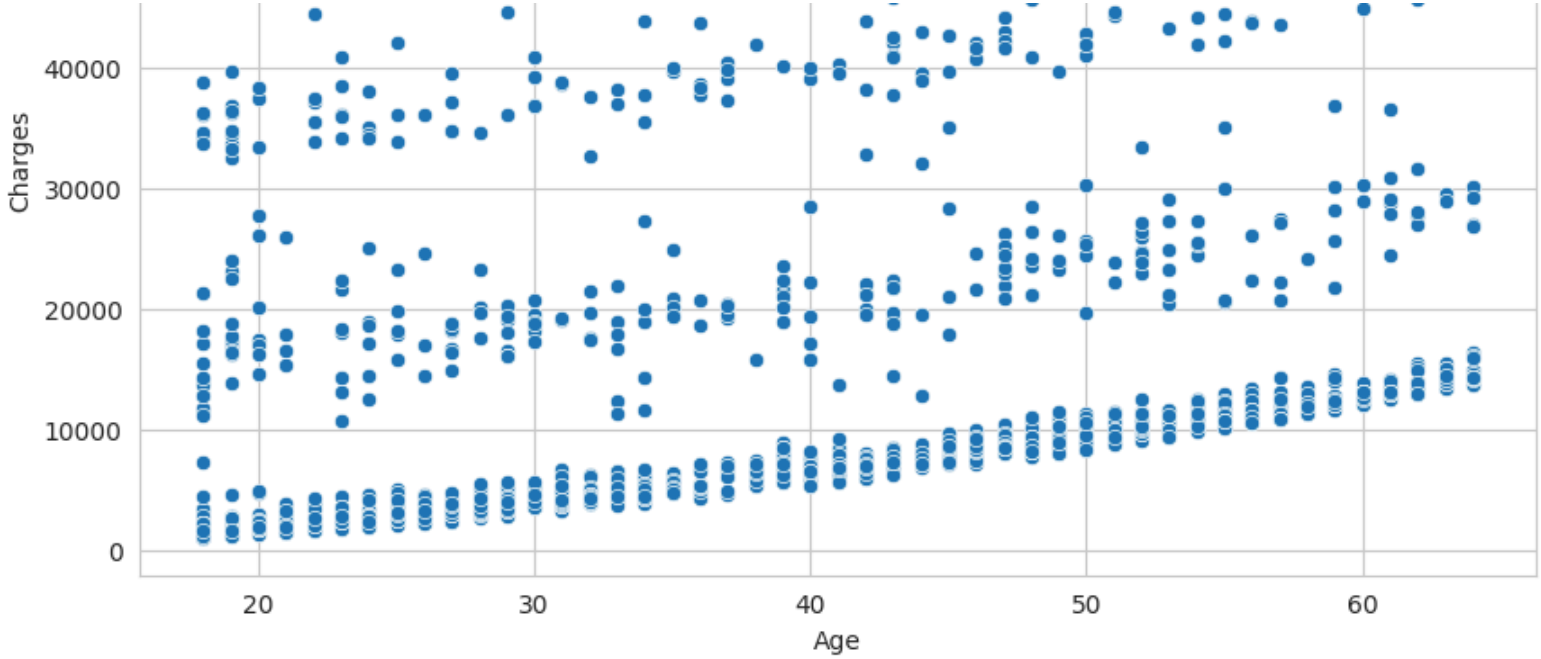
```
# Remove duplicate rows
df_insurance.drop_duplicates(inplace=True)
print('*** Duplicate Rows After Removal ***')
print(df_insurance.duplicated().sum())
print('\n')

import matplotlib.pyplot as plt
import seaborn as sns

# Set style for plots
sns.set_style('whitegrid')

# 3. Perform EDA

# a) Distribution of charges
plt.figure(figsize=(10, 6))
sns.histplot(df_insurance['charges'], kde=True, bins=30)
plt.title('Distribution of Medical Charges')
plt.xlabel('Charges')
plt.ylabel('Frequency')
plt.show()

# b) Scatter plot: age vs charges
plt.figure(figsize=(10, 6))
```

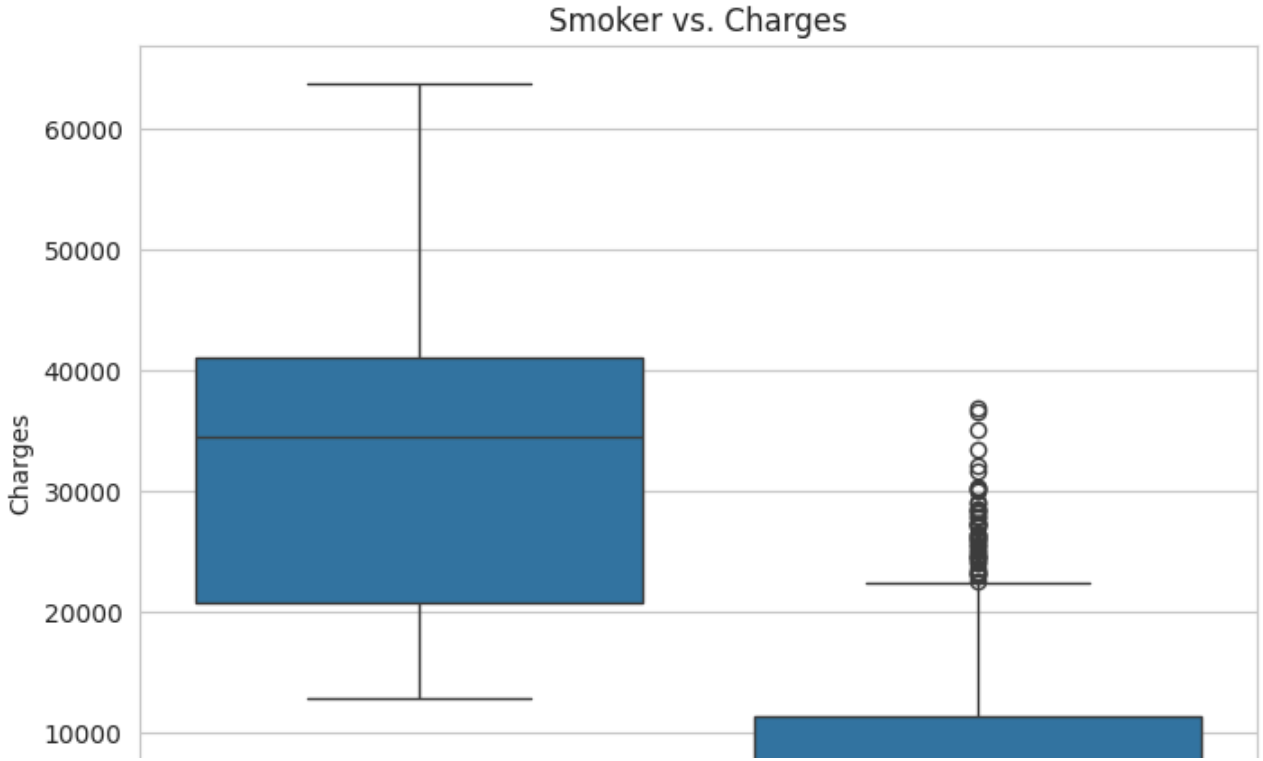
```
sns.scatterplot(x='age', y='charges', data=df_insurance)
plt.title('Age vs. Charges')
plt.xlabel('Age')
plt.ylabel('Charges')
plt.show()

# c) Boxplot: smoker vs charges
plt.figure(figsize=(8, 6))
sns.boxplot(x='smoker', y='charges', data=df_insurance)
plt.title('Smoker vs. Charges')
plt.xlabel('Smoker')
plt.ylabel('Charges')
plt.show()

# d) Correlation heatmap (first, encode categorical variables for correlation)
# Create a copy to avoid modifying the original dataframe for EDA only
df_eda = df_insurance.copy()
df_eda['sex'] = df_eda['sex'].astype('category').cat.codes
df_eda['smoker'] = df_eda['smoker'].astype('category').cat.codes
df_eda['region'] = df_eda['region'].astype('category').cat.codes

plt.figure(figsize=(10, 8))
sns.heatmap(df_eda.corr(), annot=True, cmap='coolwarm', fmt='.2f')
plt.title('Correlation Heatmap')
plt.show()
```

```
*** Duplicate Rows After Removal ***
0
```

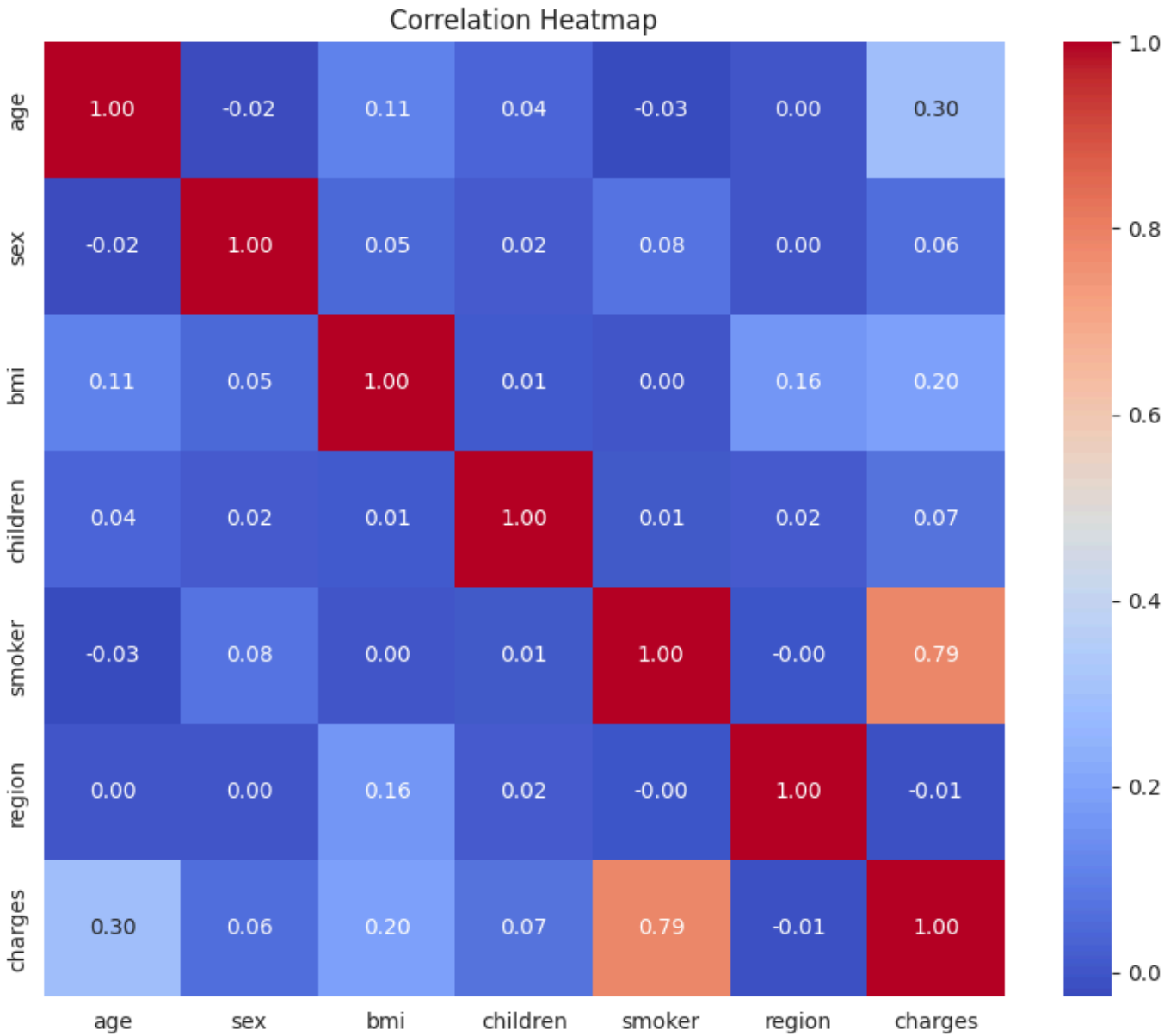





Charges
40000
30000
20000
10000
0
20
30
40
50
60
Age


Smoker vs. Charges
Charges
60000
50000
40000
30000
20000
10000

0
yes
no
Smoker


Correlation Heatmap
age 1.00 -0.02 0.11 0.04 -0.03 0.00 0.30
sex -0.02 1.00 0.05 0.02 0.08 0.00 0.06
bmi 0.11 0.05 1.00 0.01 0.00 0.16 0.20
children 0.04 0.02 0.01 1.00 0.01 0.02 0.07
smoker -0.03 0.08 0.00 0.01 1.00 -0.00 0.79
region 0.00 0.00 0.16 0.02 -0.00 1.00 -0.01
charges 0.30 0.06 0.20 0.07 0.79 -0.01 1.00
age sex bmi children smoker region charges
1.0
0.8
0.6
0.4
0.2
0.0

Section 2 – Data Preprocessing

5. Encode categorical variables properly.

```
# Identify categorical columns
categorical_cols = ['sex', 'smoker', 'region']

# Apply one-hot encoding to categorical columns
df_encoded = pd.get_dummies(df_insurance, columns=categorical_cols, drop_first=True)

print('*** DataFrame after One-Hot Encoding ***')
display(df_encoded.head())
print(f"New shape of DataFrame: {df_encoded.shape}")
```

```
*** DataFrame after One-Hot Encoding ***
```

	age	bmi	children	charges	sex_male	smoker_yes	region_northwest	region_southeast	region_southwest
0	19	27.900	0	16884.92400	False	True	False	False	True
1	18	33.770	1	1725.55230	True	False	False	True	False
2	28	33.000	3	4449.46200	True	False	False	True	False
3	33	22.705	0	21984.47061	True	False	True	False	False
4	32	28.880	0	3866.85520	True	False	True	False	False

```
New shape of DataFrame: (1337, 9)
```

6. Separate features (X) and target (y).

```
# Separate features (X) and target (y)
X = df_encoded.drop('charges', axis=1)
y = df_encoded['charges']

print('Shape of X:', X.shape)
print('Shape of y:', y.shape)

print('\nFirst 5 rows of X:')
display(X.head())
```

```
print('\nFirst 5 values of y:')
display(y.head())
```

```
Shape of X: (1337, 8)
Shape of y: (1337,)

First 5 rows of X:
```

	age	bmi	children	sex_male	smoker_yes	region_northwest	region_southeast	region_southwest
0	19	27.900	0	False	True	False	False	True
1	18	33.770	1	True	False	False	True	False
2	28	33.000	3	True	False	False	True	False
3	33	22.705	0	True	False	True	False	False
4	32	28.880	0	True	False	True	False	False

```
First 5 values of y:
```

	charges
0	16884.92400
1	1725.55230
2	4449.46200
3	21984.47061
4	3866.85520

dtype: float64

7. Perform train-test split (80-20).

```
from sklearn.model_selection import train_test_split

# Perform train-test split (80-20)
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

print('Shape of X_train:', X_train.shape)
```

```
print('Shape of X_test:', X_test.shape)
print('Shape of y_train:', y_train.shape)
print('Shape of y_test:', y_test.shape)
```

```
Shape of X_train: (1069, 8)
Shape of X_test: (268, 8)
Shape of y_train: (1069,)
Shape of y_test: (268,)
```

8. Apply feature scaling (if required).

```
from sklearn.preprocessing import StandardScaler

# Initialize the StandardScaler
scaler = StandardScaler()

# Apply scaling only to numerical features in X_train and X_test
# We exclude the one-hot encoded boolean columns for sex, smoker, and region
numerical_cols = ['age', 'bmi', 'children']

X_train[numerical_cols] = scaler.fit_transform(X_train[numerical_cols])
X_test[numerical_cols] = scaler.transform(X_test[numerical_cols])

print('*** X_train after scaling (first 5 rows) ***')
display(X_train.head())
print('\n*** X_test after scaling (first 5 rows) ***')
display(X_test.head())
```

```
*** X_train after scaling (first 5 rows) ***
```

	age	bmi	children	sex_male	smoker_yes	region_northwest	region_southeast	region_southwest
1114	-1.157680	-0.996928	-0.907908	True	False	False	False	False
968	-1.300619	-0.792762	0.766904	True	False	False	False	False
599	0.914926	1.154664	0.766904	False	False	True	False	False
170	1.701087	1.806837	-0.907908	True	False	False	True	False
275	0.557580	-0.651417	0.766904	False	False	False	False	False

```
*** X_test after scaling (first 5 rows) ***
```

	age	bmi	children	sex_male	smoker_yes	region_northwest	region_southeast	region_southwest
900	0.700518	-1.326734	-0.907908	True	False	False	False	False
1064	-0.728865	-0.816733	2.441716	False	False	False	False	True
1256	0.843457	0.966203	1.604310	False	False	True	False	False
298	-0.585927	0.636397	1.604310	True	True	True	False	False
237	-0.585927	1.297663	0.766904	True	False	False	True	False

9. Explain why scaling is / isn't required.

Why Scaling is Required for this dataset:

Feature scaling is important for this dataset because of the varying scales of our numerical features: 'age', 'bmi', and 'children'.

1. **Varying Ranges:** 'Age' typically ranges from 18-64, 'BMI' from 16-53, and 'Children' from 0-5. These features have different magnitudes and ranges.
2. **Algorithm Performance:** Machine learning algorithms, particularly those that use distance calculations (like k-nearest neighbors, support vector machines) or gradient descent (like Linear Regression), are sensitive to the scale of input features. If features are not scaled, features with larger values might dominate the distance calculations or have a disproportionately large impact on the loss function, leading to a suboptimal model.
3. **Linear Regression:** While not strictly necessary for basic Linear Regression (as coefficients can adjust to different scales), scaling can lead to faster convergence during optimization and can help in interpreting the relative importance of features if the coefficients

are compared. More importantly, it becomes crucial if regularization techniques (like Ridge or Lasso) are applied, as these methods penalize large coefficients.

In this case, `StandardScaler` has been used to transform the numerical features so that they have a mean of 0 and a standard deviation of 1. This ensures that all numerical features contribute equally to the model, preventing any single feature from dominating due to its scale. Categorical features created by one-hot encoding (boolean values) do not require scaling.

Section 3 – Model Building

10. Train a Linear Regression model.

```
from sklearn.linear_model import LinearRegression

# Initialize the Linear Regression model
linear_reg_model = LinearRegression()

# Train the model on the scaled training data
linear_reg_model.fit(X_train, y_train)

print("Linear Regression model trained successfully.")
```

```
Linear Regression model trained successfully.
```

Section 4 – Model Evaluation

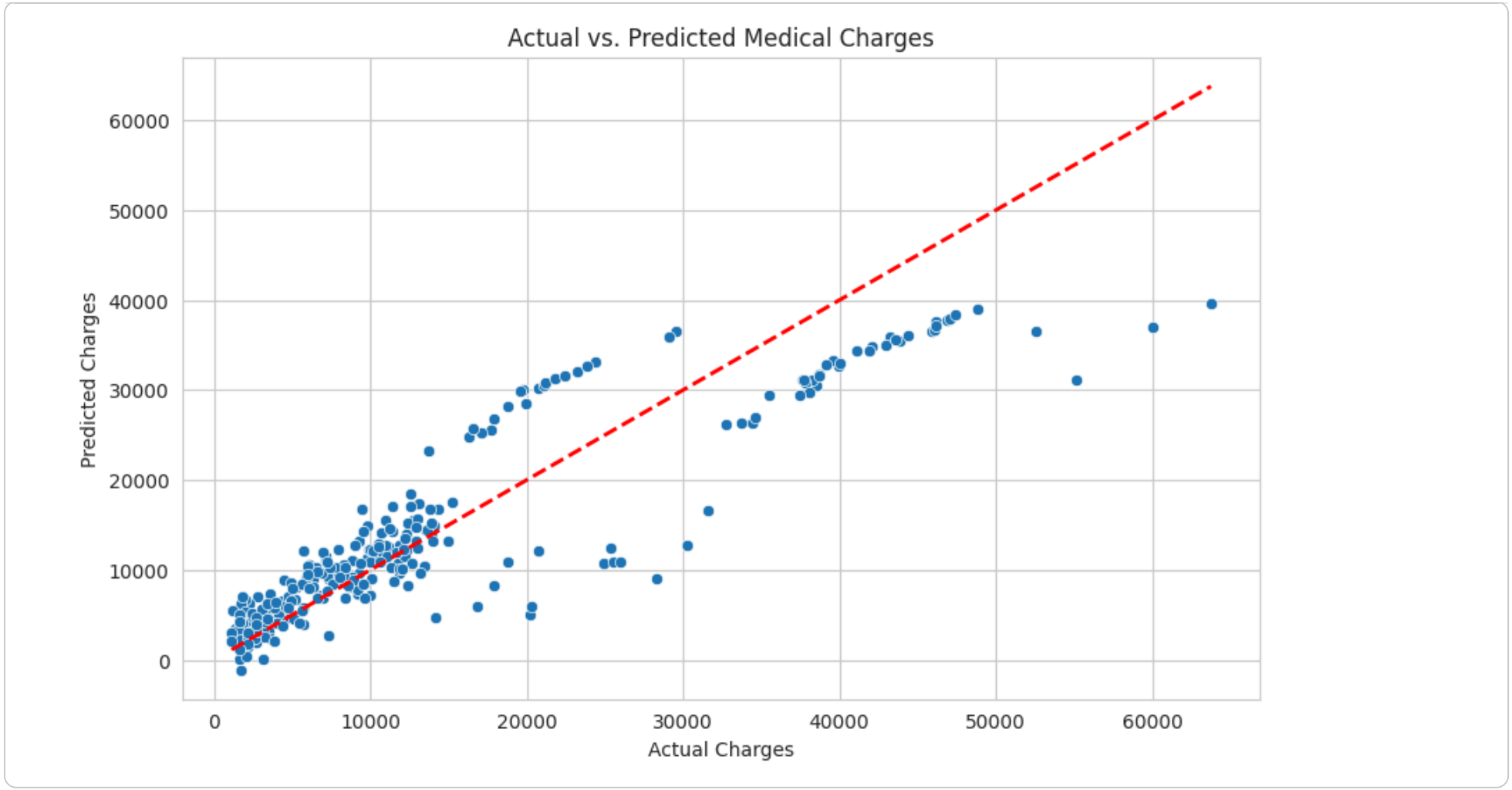
```
from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score
import numpy as np

# Make predictions on the test data
y_pred = linear_reg_model.predict(X_test)

# Calculate evaluation metrics
mae = mean_absolute_error(y_test, y_pred)
mse = mean_squared_error(y_test, y_pred)
rmse = np.sqrt(mse)
r2 = r2_score(y_test, y_pred)
```

```
print(f'Mean Absolute Error (MAE): {mae:.2f}')
print(f'Mean Squared Error (MSE): {mse:.2f}')
print(f'Root Mean Squared Error (RMSE): {rmse:.2f}')
print(f'R-squared (R²): {r2:.2f}')
```

```
Mean Absolute Error (MAE): 4177.05
Mean Squared Error (MSE): 35478020.68
Root Mean Squared Error (RMSE): 5956.34
R-squared (R²): 0.81
```

Plotting Actual vs. Predicted Values

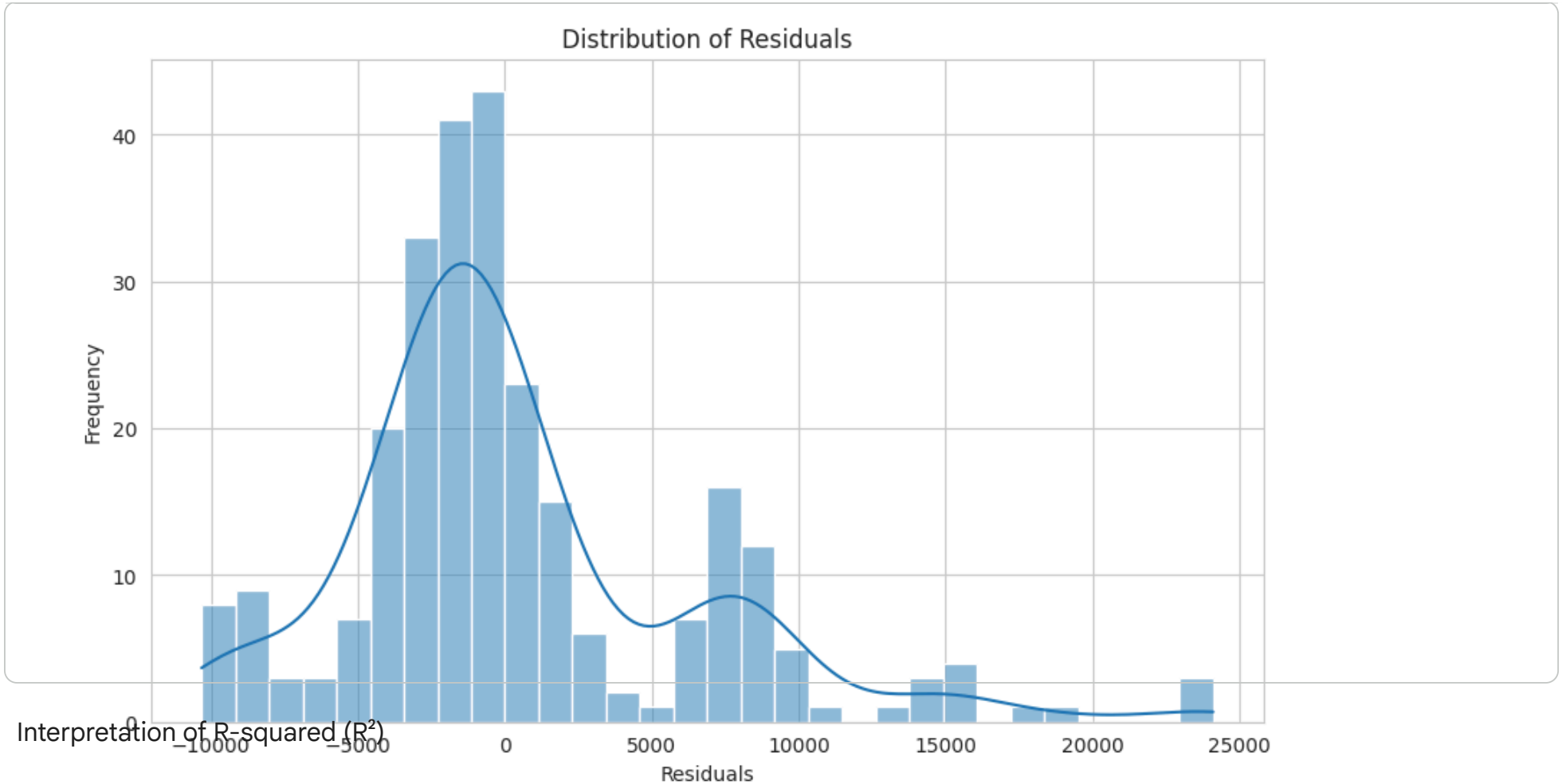
```
plt.figure(figsize=(10, 6))
sns.scatterplot(x=y_test, y=y_pred)
plt.plot([y_test.min(), y_test.max()], [y_test.min(), y_test.max()], 'r--', lw=2)
plt.xlabel('Actual Charges')
plt.ylabel('Predicted Charges')
plt.title('Actual vs. Predicted Medical Charges')
plt.show()
```



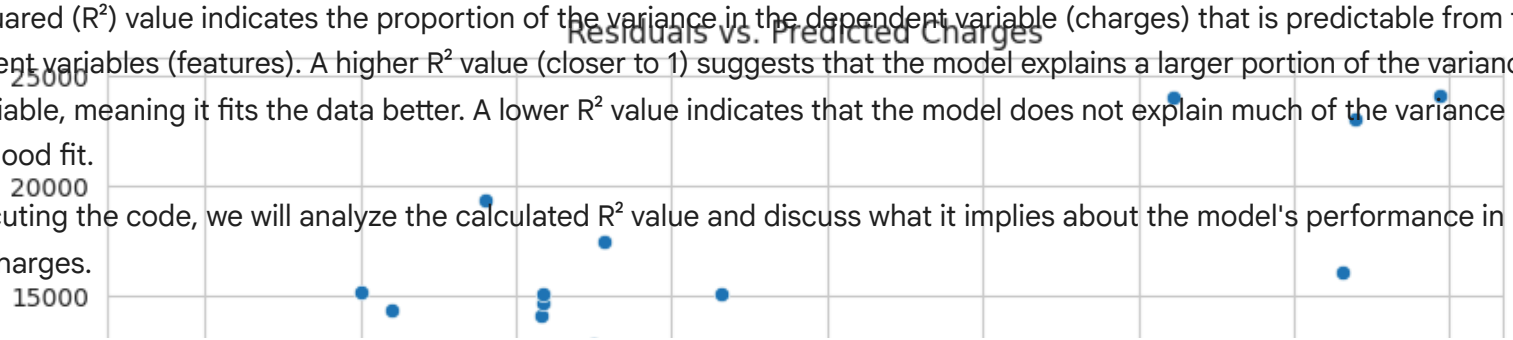
Plotting Residuals

```
residuals = y_test - y_pred

plt.figure(figsize=(10, 6))
sns.histplot(residuals, kde=True, bins=30)
plt.xlabel('Residuals')
plt.ylabel('Frequency')
plt.title('Distribution of Residuals')
plt.show()
```

```
plt.figure(figsize=(10, 6))
sns.scatterplot(x=y_pred, y=residuals)
plt.axhline(y=0, color='r', linestyle='--')
plt.xlabel('Predicted Charges')
plt.ylabel('Residuals')
plt.title('Residuals vs. Predicted Charges')
plt.show()
```



Interpretation of R-squared (R^2)

The R-squared (R^2) value indicates the proportion of the variance in the dependent variable (charges) that is predictable from the independent variables (features). A higher R^2 value (closer to 1) suggests that the model explains a larger portion of the variance in the target variable, meaning it fits the data better. A lower R^2 value indicates that the model does not explain much of the variance and might not be a good fit.

After executing the code, we will analyze the calculated R^2 value and discuss what it implies about the model's performance in predicting medical charges.